

Design Verification Plan & Report (Virtual Test) — VBF-T5R3-DVPR-01

****Case**:** t5_r3 · ****Design under verification**:** Y4 (primary, `cell/params\_y4.json`) and Y3 (alternate, `cell/params\_y3.json`) · ****Date**:** 2026-08-26 · ****Scope**:** simulation-based verification only; uncovered conditions marked N/A honestly.

Verification Results — Y4 (Primary)

Item	Condition	Result value	Determination	Source
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1C discharge capacity	1C CC discharge, 298.15 K, full-order DFN	5.0442 Ah	No capacity threshold in contract (information)	`cell/r6_y4_1c_dfn.json:capacity_ah`
Rated energy	same	18.2855 Wh	Information	`cell/r6_y4_energy_dfn.json:energy_wh`
Energy density (gravimetric)	25 °C 1C, contract formula	535.72 Wh/kg	(PASS) PASS vs >= 500.94 Wh/kg	`cell/r6_y4_energy_dfn.json:energy_density_wh_kg`
Volumetric energy density	same	995.79 Wh/L	Information	`cell/r6_y4_energy_dfn.json:energy_density_wh_l`
4C fast-charge temperature rise	4C CC charge, 318.15 K ambient, lumped thermal, DFN	T_max = 327.60 K (54.4 °C)	(PASS) PASS vs <= 333.15 K	`cell/r6_y4_4c_dfn.json:T_max_K`
4C fast-charge plating	same, `--plating`; anode surface potential	min +0.0200 V (never < 0 V)	(PASS) PASS (plated = false)	`cell/r6_y4_4c_dfn.json:anode_potential_v`
4C CC charge window	20 A CC from 2.5 V to 4.2 V cut-off	3.48 Ah (~69 % SOC) in 626 s	Information (task = support 4C without plating; CV tail not simulated)	`cell/r6_y4_4c_dfn.json:time_s/voltage_v`
Voltage window	parameter set limits	2.5 – 4.2 V	Information	`param_dump.txt`

Verification Results — Y3 (Supply-friendly alternate, Al 8 µm / sep 9 µm)

Item	Condition	Result value	Determination	Source
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Energy density	25 °C 1C, contract formula	526.24 Wh/kg	(PASS) PASS vs >= 500.94 Wh/kg	`cell/r6_y3_energy_dfn.json:energy_density_wh_kg`
4C fast-charge temperature rise	lumped thermal, DFN	T_max = 329.38 K	(PASS) PASS vs <= 333.15 K	`cell/r6_y3_4c_dfn.json:T_max_K`
4C fast-charge plating	`--plating`, DFN	min +0.0162 V	(PASS) PASS (plated = false)	`cell/r6_y3_4c_dfn.json:anode_potential_v`
1C capacity	DFN	5.0442 Ah	Information	`cell/r6_y3_1c_dfn.json:capacity_ah`

Items Explicitly N/A (beyond pure simulation boundary, requires physical experiment)

Item	Status
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Nail penetration	N/A (requires physical experiment)
Overcharge to thermal runaway	N/A (requires physical experiment)
Crush / drop	N/A (requires physical experiment)
Cycle life	N/A (no aging model run; task has no durability metric — must not fabricate)
Rate-pulse internal resistance	N/A (parameter-caliber DCR 2.723 mΩ available as reference: `r6_y4_energy_dfn.json:dc_r_ohm`)
Enclosure / tabs / manufacturing line validation	N/A (beyond pure-simulation boundary)

Conclusion

****All contractual criteria PASS at full-order DFN precision.**** Pass criteria: energy density >= 500.94 Wh/kg (PASS) (535.72), 4C fast charge without lithium plating (PASS) (min anode potential +0.0200 V), maximum temperature <= 60 °C (PASS) (327.60 K). Plating margin is thin by design trade (kinetic warmth ↔ temperature) — recommend physical-cell validation

sequence: foil handling (5 μm Cu handling sensitivity), 4C CC charging anode reference-electrode test, and CV-tail confirmation before productization. Uncovered items listed above are the paper limitations.